

Level 2 Sliding Window Problems

1. Longest Substring Without Repeating Characters

→ Variable Window + HashMap

```
s = "abcabcbb"

def subarr_longest(s):

    last_seen = {}

    left = 0
    max_length = 0

    for right in range(len(s)):

        char = s[right]

        if char in last_seen:
            left = max(left, last_seen[char] + 1)

        last_seen[char] = right
        current_length = right - left + 1

        max_length = max(max_length, current_length)

    return max_length

print(subarr_longest(s))
```

2. Longest Subarray with Sum $\leq K$

→ Variable Window

```
nums = [1, 2, 1, 0, 1, 1, 0]
k = 4

def long_subarr(nums, k):
    left = 0
    max_len = 0
    wind_sum = 0

    for right in range(len(nums)):

        wind_sum += nums[right]

        while wind_sum > k:
            wind_sum -= nums[left]
            left += 1

        cur_len = right - left + 1

        max_len = max(max_len, cur_len)

    return max_len

print(long_subarr(nums,k))
```

3. Minimum Size Subarray Sum

→ Expand + Shrink

```
target = 7
nums = [2,3,1,2,4,3]

def min_subarr_len(target, nums):

    left = 0
    wind_sum = 0
    min_len = float('inf')

    for right in range(len(nums)):

        wind_sum += nums[right]

        while wind_sum >= target:
            curr_len = right - left + 1
            min_len = min(min_len, curr_len)
            wind_sum -= nums[left]
            left += 1

    if min_len == float('inf'):
        return 0

    return min_len

print(min_subarr_len(target,nums))
```